

Contrastive Learning for Visual Representations

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Abstract

We propose a contrastive self-supervised learning framework for visual representations, achieving 85.4% linear evaluation accuracy on ImageNet with a ResNet-50 backbone and no labels during pre-training.

1 Introduction

Self-supervised learning has achieved remarkable progress in recent years. SimCLR (Chen et al., 2020) learns representations by maximising agreement between augmented views of the same image using a contrastive loss. MoCo (He et al., 2020) maintains a memory bank of negative keys to enable large-batch training without a correspondingly large batch size.

2 Experiments

We pre-train on ImageNet-1K for 200 epochs using two 180-degree rotations, colour jitter, and Gaussian blur as augmentations.

Linear evaluation follows the standard protocol from SimCLR.

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References

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